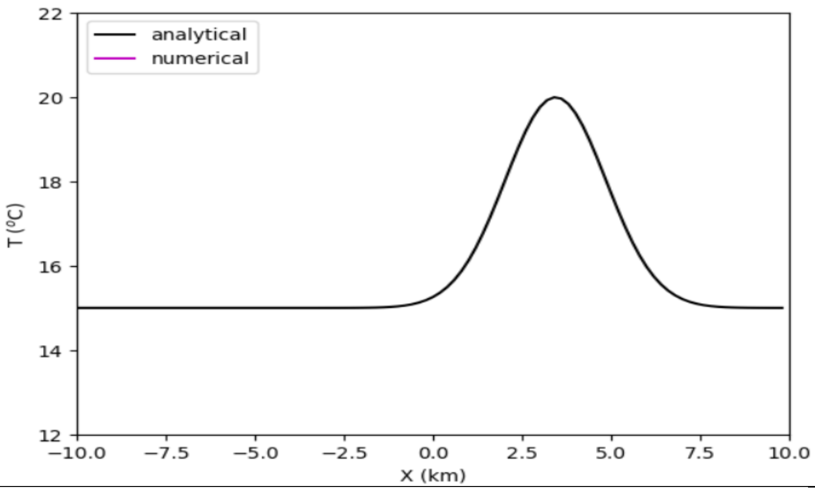
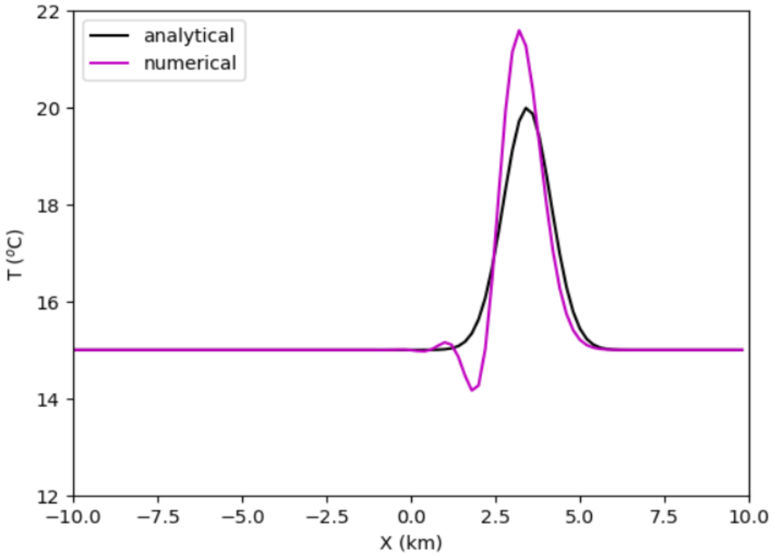
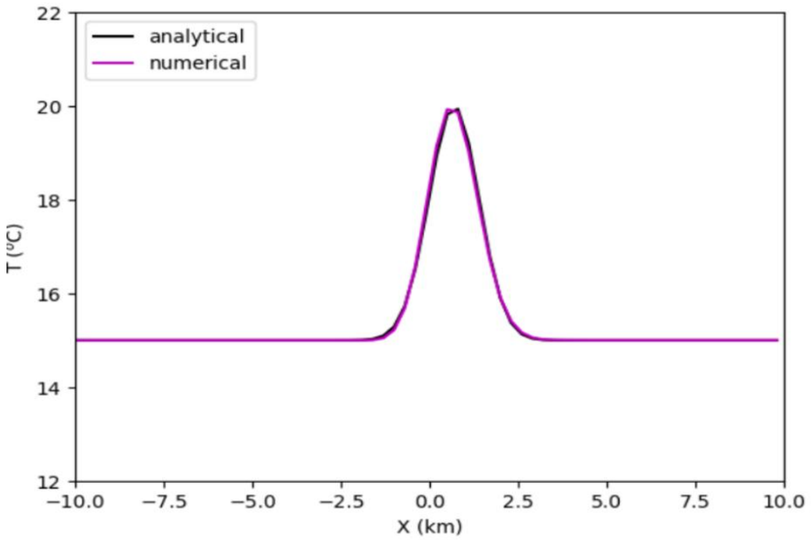
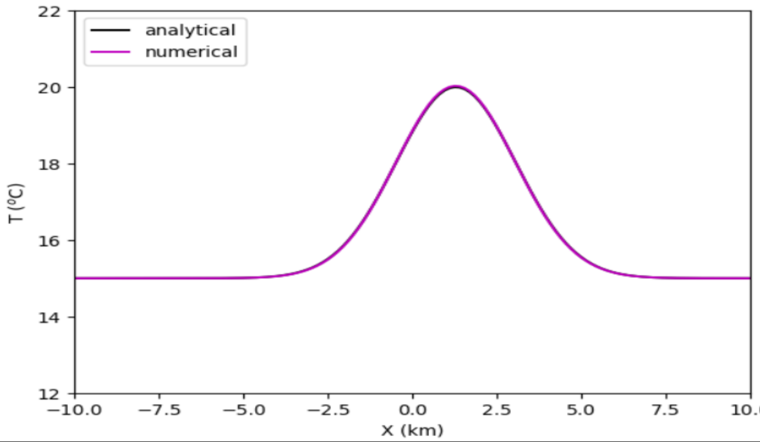
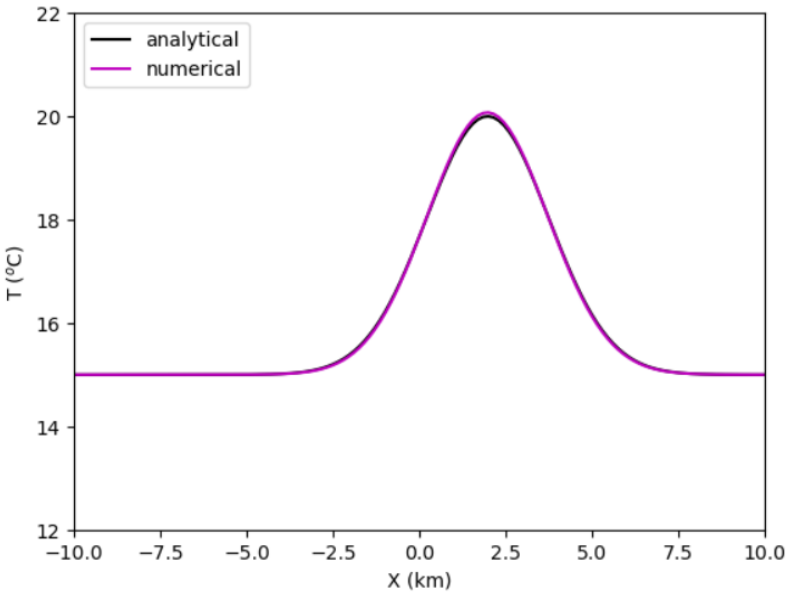
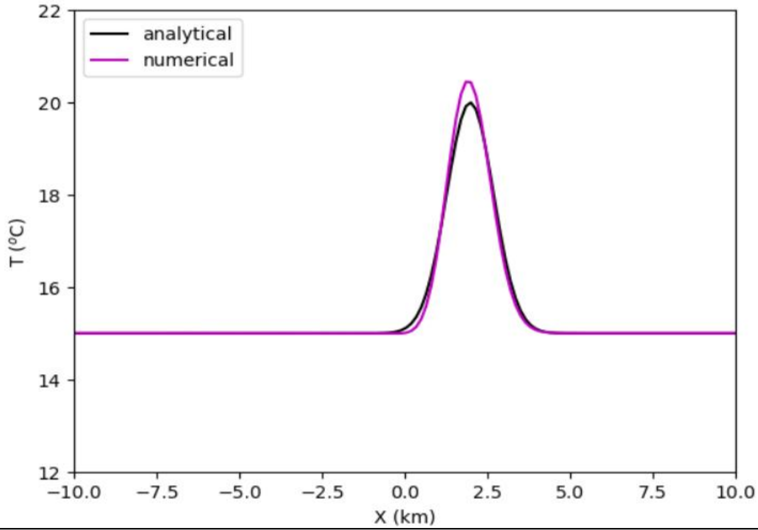
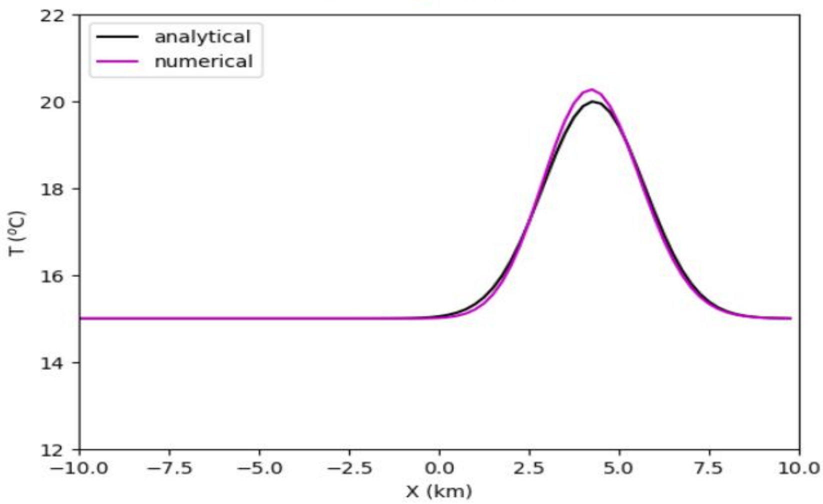
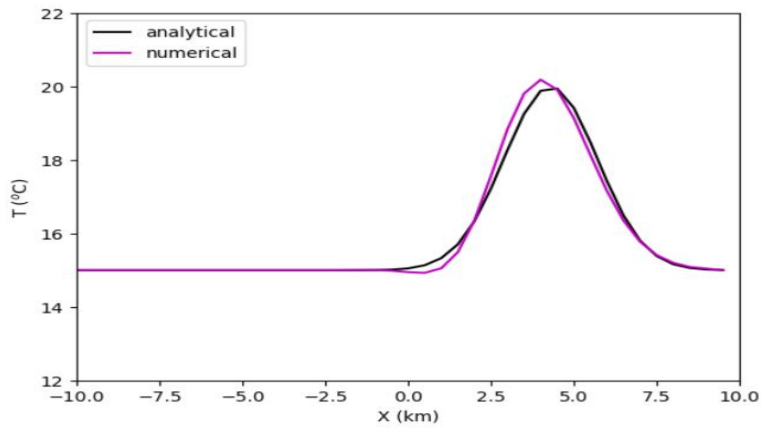
Exercise 10

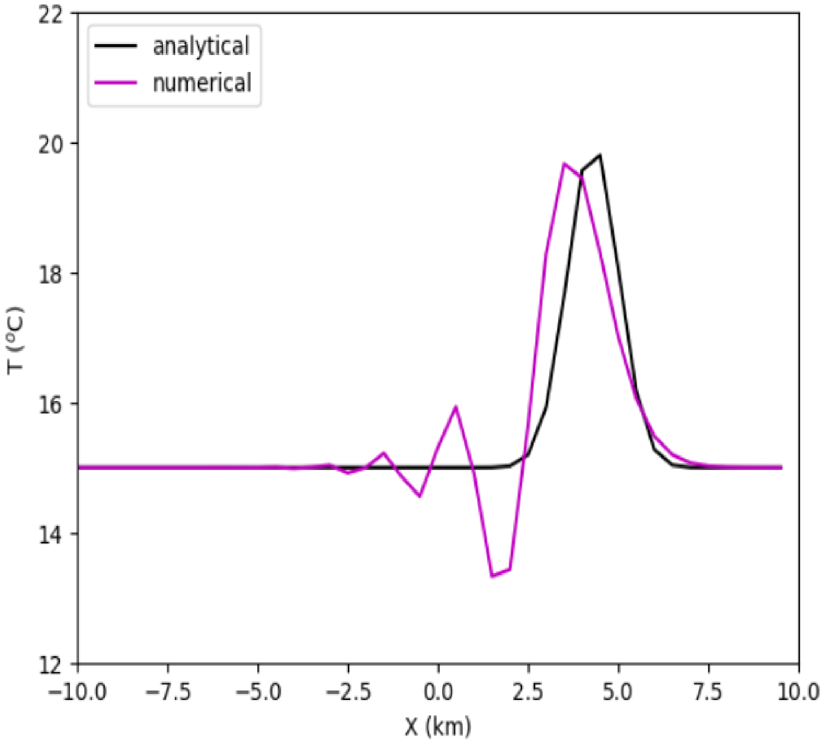
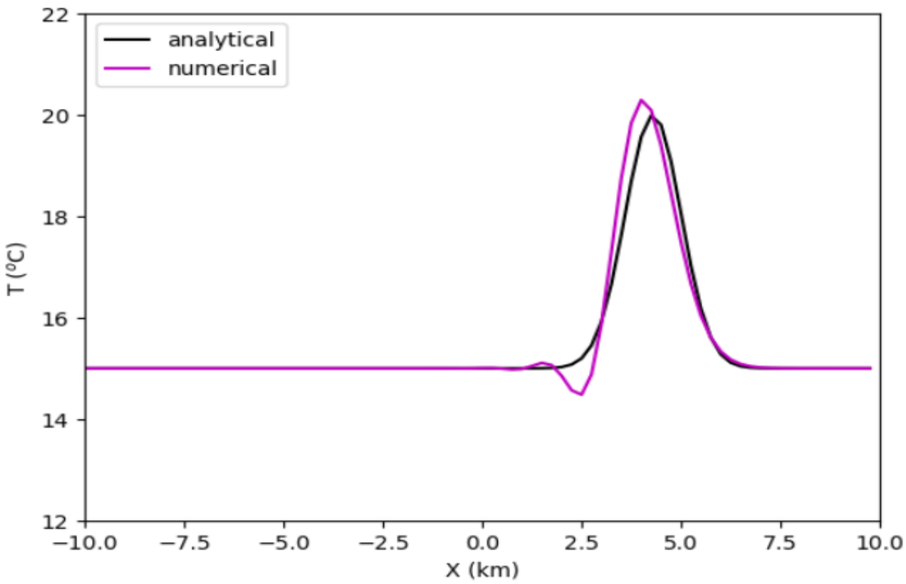
Space-time parameters	Plots results
Grid steps : 100 Number of time steps : 87 Timestep = 200 s Max numerical speed DX/DT = 1.0 m/s Physical speed = 0.2 m/s Courant number = 0.2	
Grid steps : 100 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 1.0 m/s Physical speed = 0.4 m/s Courant number = 0.4	
Grid steps : 67 Number of time steps : 87 Timestep = 100 s Max numerical speed DX/DT = 3.0 m/s Physical speed = 0.08 m/s Courant number = 0.02666666666666667	

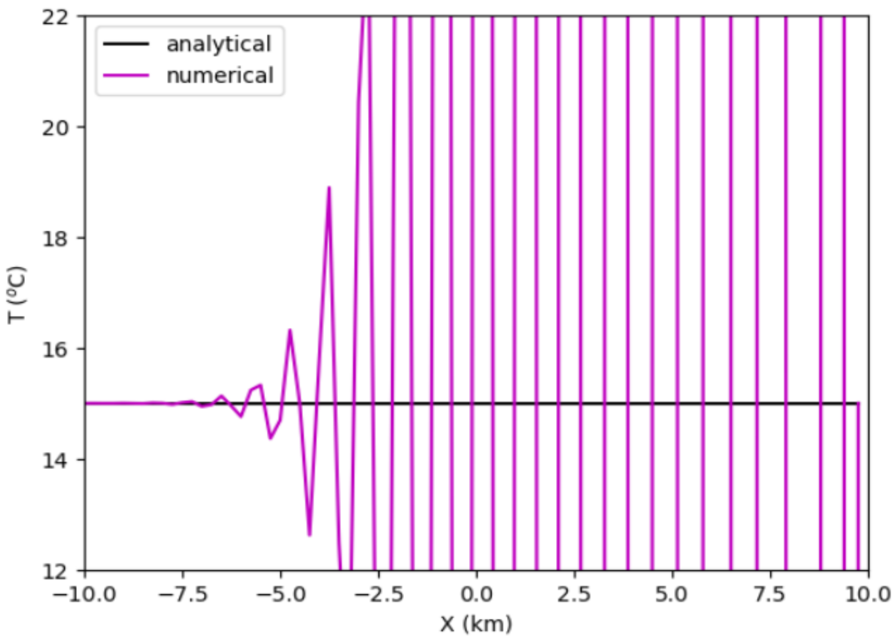
<table>
<tr>
<td>Grid steps : 134
Number of time steps : 44
Timestep = 200 s
Max numerical speed
DX/DT = 0.75 m/s
Physical speed = 0.15 m/s
Courant number = 0.2</td>
<td>
</td>
</tr>
<tr>
<td>Grid steps : 134
Number of time steps : 44
Timestep = 200 s
Max numerical speed
DX/DT = 0.75 m/s
Physical speed = 0.23 m/s
Courant number =
0.30666666666666664
Increased the parameter σ:
2500</td>
<td>
</td>
</tr>
<tr>
<td>decrease the parameter σ:
1000
Grid steps : 134
Number of time steps : 44
Timestep = 200 s
Max numerical speed
DX/DT = 0.75 m/s
Physical speed = 0.23 m/s
Courant number =
0.30666666666666664</td>
<td>
</td>
</tr>
</table>

Exercise 11- question 5

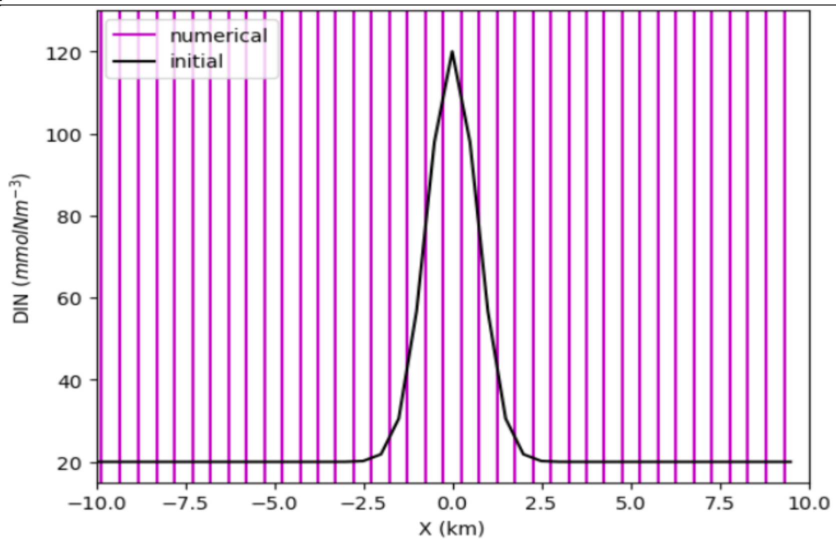
Space-time parameters	Plots results
Grid steps : 80 Number of time steps : 87 Timestep = 100 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 0.5 m/s Courant number = 0.2	
Grid steps : 40 Number of time steps : 87 Timestep = 100 s Max numerical speed DX/DT = 5.0 m/s Physical speed = 0.5 m/s Courant number = 0.1	

Question 6

<table>
<tr>
<td>If c=0.5, dx=500,
dt=100,
sigma=1000
Grid steps : 40
Number of time
steps : 87
Timestep = 100 s
Max numerical
speed DX/DT = 5.0
m/s
Physical speed =
0.5 m/s
Courant number =
0.1</td>
<td>
</td>
</tr>
<tr>
<td>Grid steps : 80
Number of time
steps : 173
Timestep = 50 s
Max numerical
speed DX/DT = 5.0
m/s
Physical speed =
0.5 m/s
Courant number =
0.1</td>
<td>This plot shows when to recover the same degree of accuracy of the previous solution.

</td>
</tr>
</table>

Grid steps : 80 Number of time steps : 173 Timestep = 50 s Max numerical speed DX/DT = 5.0 m/s Physical speed = 2 m/s Courant number = 0.4	

Exercise 12 (DIFFUSION)

Grid steps : 40 Number of time steps : 104 Timestep = 100 s Max numerical diffusion DX^2/DT = 2500.0 m2/s Physical diffusion = 2000.0 m2/s Courant number (diffusion) = 0.8 If kappa=2000 and dt=100	

Grid steps : 40 Number of time steps : 208 Timestep = 50 s Max numerical diffusion DX^2/DT = 5000.0 m2/s Physical diffusion = 2000.0 m2/s Courant number (diffusion) = 0.4 If dt=50